



Hospital and long-term results of carotid endarterectomy in patients with different severity of coronary artery lesion according to syntax score

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SUMMARY

Aim: Analysis of in-hospital and long-term results of carotid endarterectomy (CEE) in patients with different severity of coronary atherosclerosis.

Material and methods: This comparative, retrospective, open study for the period from January 2013 to April 2020 included 1719 patients operated on for occlusive-stenotic lesions of the internal carotid arteries (ICA). Classical and eversion CEA were used as revascularization strategies. The criteria for inclusion in the study were: 1. Presence of coronary angiography within six months before the present CEE; 2. A history of myocardial revascularization in patients with severe coronary lesions. Depending on the severity of coronary atherosclerosis, all patients were divided into 3 groups: Group 1–871 (50.7 %) patients – with the presence of hemodynamically significant stenosis of the coronary arteries (CA) with a history of myocardial revascularization; Group 2–496 (28.8 %) patients – with the presence of hemodynamically insignificant lesions of

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the coronary artery (up to 70 %, not inclusive, and the trunk of the left coronary artery, up to 50 %, not inclusive); Group 3–352 (20.5 %) patients – without signs of atherosclerotic lesions of the coronary artery. In group 1, the observation period was 56.8 ± 23.2 months, in group 2– 62.0 ± 15.6 months, in group 3– 58.1 ± 20.4 months.

Results: During the hospital observation period, there were no significant intergroup differences in the number of complications. All cardiovascular events were detected in isolated cases. The most common injury was damage to the cranial nerves, diagnosed in every fifth patient in the total sample. The combined endpoint (CET), including death + myocardial infarction (MI) + acute cerebrovascular accident/transient ischemic attack (stroke/TIA), was 0.75 % ($n=13$). In the long-term follow-up period, when comparing survival curves, group 3 revealed the largest number of ischemic strokes ($p = 0.007$), myocardial infarction ($p = 0.03$), and CCT ($p = 0.005$). There were no intergroup differences in the number of deaths ($p=0.62$).

Conclusion: The results of the study showed that there was no significant intergroup difference in the development of complications at the hospital postoperative stage. However, in the long-term follow-up period, a group of patients with isolated lesions of the ICA demonstrated a rapid increase in the number of MI, stroke/TIA, and a combined endpoint, which was apparently associated with low compliance and progression of atherosclerosis in previously unaffected arteries.

Introduction

Speaking about patients who are referred for carotid endarterectomy (CEE), we must remember that not all of them have isolated stenoses of the internal carotid arteries (ICA).^{1–5} According to a number of studies, in a significant proportion of patients requiring CEA, coronary lesions are visualized, which raises the question of the need for myocardial revascularization.^{6–10} How dangerous it is to perform CEE in a sample of patients with severe coronary atherosclerosis relative to those who have differentiated insignificant stenoses of the coronary arteries, according to current recommendations, is determined by a multidisciplinary council personalized for each patient (class 1, level of evidence C).^{1,5}

The existing problems in the evidence base do not demonstrate a complete picture of the risks when performing CEA in patients with varying severity of coronary atherosclerosis.^{11–15} It is necessary to understand whether any additional measures are needed to prevent postoperative complications or whether a standard management protocol is sufficient. The risks of possible cardiovascular events in the long-term period have not been fully determined, as well as the factors that influenced its severity in different samples of patients.¹

The purpose of this work was to analyze in-hospital and long-term postoperative complications after CEA in patients with different severity of coronary atherosclerosis according to SYNTAX SCORE.

Material and methods

This comparative retrospective open study for the period from January 2013 to April 2020 included 1719 patients operated on for occlusive-stenotic lesions of the ICA. Classical and eversion CEA were used as revascularization strategies.

Depending on the severity of coronary atherosclerosis, all patients were divided into 3 groups:

Group 1–871 (50.7 %) patients with hemodynamically significant stenoses of the coronary arteries (CA) with a history of myocardial revascularization (coronary artery bypass grafting (CABG) or percutaneous coronary intervention (PCI));

Group 2–496 patients (28.8 %) with hemodynamically insignificant lesions of the coronary artery (up to 70 % inclusive and the trunk of the left coronary artery up to 50 % inclusive);

Group 3–352 patients (20.5 %) without signs of atherosclerotic lesions of the coronary artery.

The patient's condition was monitored by returning the patient to the clinic every six months. In group 1, the observation period was 56.8 ± 23.2 months, in group 2– 62.0 ± 15.6 months, in group 3– 58.1 ± 20.4 months.

Assessment of the severity of coronary atherosclerosis

The severity of coronary atherosclerosis was calculated using the interactive SYNTAX Score calculator (www.syntaxscore.com). According to the severity of the lesion, based on this calculator, the following gradation is distinguished: low level of lesion (22 points or less), intermediate (23–32 points) and severe (33 points or more). In group 1 of patients with severe coronary lesions and a history of myocardial revascularization, residual SYNTAX was calculated, reflecting the significance of residual coronary artery stenoses.

Instrumental studies

To conclude about the presence of multifocal atherosclerosis at the preoperative stage, the patient underwent screening color duplex scanning (CDS) of the brachiocephalic bed, arteries of the lower extremities, aortic arch (using a linear sensor with a frequency of 7–7.5 MHz), heart (using a sector sensor with a frequency 2.5–4 MHz) using Acuson 128XP (Acuson, USA) and Sonos 2500 (HewlettPackard, USA) devices. Coronary angiography was performed using an Innova 2100 angiographic unit (General Electric,

Table 1
Clinical and demographic characteristics, neurological status.

Index	Group 1 (CEE with hemodynamically significant coronary artery stenoses), n=871	Group 2 (CEE with hemodynamically insignificant coronary artery stenoses), n=496	Group 3 (CEA without coronary artery stenosis), n=352	p
Age	66,3±5,5	64,9±5,8	62,7±4,8	0,35
Male	583 (66,9)	305 (61,5)	197 (55,9)	P1–2: 0,13 P1–3: 0,001 P2–3: 0,30
Angina pectoris I–II functional class	871 (100)	54 (10,8)	0	P1–2: 0,0001 P1–3: 0,0001 P2–3: 0,0052
History of myocardial infarction	248 (28,4)	0	0	P1–2: 0,0001 P1–3: 0,0001 P2–3: 0,9999
Diabetes mellitus	25 (2,8)	12 (2,4)	8 (2,2)	P1–2: 0,9999 P1–3: 0,9999 P2–3: 0,9999
Chronic obstructive pulmonary disease	10 (1,1)	7 (1,4)	10 (2,8)	P1–2: 0,9999 P1–3: 0,9999 P2–3: 0,9999
Left ventricular ejection fraction, %	59,8±7,0	60,1±6,5	61,3±4,2	0,41
Pulmonary hypertension	3 (0,3)	0	0	P1–2: 0,9999 P1–3: 0,9999 P2–3: 0,9999
Post-infarction left ventricular aneurysm	5 (0,5)	0	0	P1–2: 0,9999 P1–3: 0,9999 P2–3: 0,9999
History of percutaneous coronary intervention	493 (56,6)	0	0	P1–2: 0,0001 P1–3: 0,0001 P2–3: 0,9999
History of coronary artery bypass surgery	378 (43,3)	0	0	P1–2: 0,0001 P1–3: 0,0001 P2–3: 0,9999
Neurological status				
NIHSS scale	1,87±0,49	1,80±0,45	1,83±0,56	0,58
Rivermead Mobility Index	13,1±2,1	12,9±1,9	13,0±1,5	0,16
Modified Rankin Scale	1,60±0,52	1,55±0,81	1,60±0,63	0,2
Barthel scale	95,4±3,1	94,5±5,0	93,1±6,7	0,45
History of ACVA/TIA	231 (26,5)	128 (25,8)	164 (46,6)	P1–2: 0,9999

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Table 1 (continued)

Index	Group 1 (CEE with hemodynamically significant coronary artery stenoses), <i>n</i> =871	Group 2 (CEE with hemodynamically insignificant coronary artery stenoses), <i>n</i> =496	Group 3 (CEA without coronary artery stenosis), <i>n</i> =352	p
				P1–3: 0,0001 P2–3: 0,0001

USA).

Inclusion/exclusion criteria

The criteria for inclusion in the study were: (1) presence of coronary angiography within six months before the present CEE; (2) a history of myocardial revascularization in patients with severe coronary lesions.

The exclusion criteria were: (1) the presence of pathology that limits follow-up of the patient in the long-term period; (2) no history of coronary angiography; (3) the need for myocardial revascularization as the next step.

Features of performing carotid endarterectomy

Protection of the brain during CEE was carried out as follows. Invasive measurement of retrograde pressure in the ICA was performed intraoperatively. After clamping the arteries, the systolic blood pressure increased to 180–220 mm Hg. Art., 5 thousand units of heparin were administered intravenously. The operation was performed under general anesthesia. A 6-0 Prolene thread was used as a suture material to perform the vascular anastomosis. Management of patients in the hospital postoperative period was carried out according to a standard scheme, including mandatory consultation with a cardiologist, otolaryngologist, and the prescription of non-steroidal anti-inflammatory and nootropic drugs.

Table 2

Angiographic and perioperative characteristics.

Index	Group 1 (CEE with hemodynamically significant coronary artery stenoses), <i>n</i> =871	Group 2 (CEE with hemodynamically insignificant coronary artery stenoses), <i>n</i> =496	Group 3 (CEA without coronary artery stenosis), <i>n</i> =352	p
Hemodynamically significant stenoses of the ICA on both sides	256 (29,3)	117 (23,5)	129 (36,6)	P1–2: 0,07 P1–3: 0,03 P2–3: 0,0001
Contralateral ICA occlusion	68 (7,8)	45 (9,0)	31 (8,8)	P1–2: 0,9999 P1–3: 0,9999 P2–3: 0,9999
Unstable atherosclerotic plaque in the ipsilateral ICA	232 (26,6)	169 (34,0)	211 (59,9)	P1–2: 0,0174 P1–3: 0,0001 P2–3: 0,0001
Unstable atherosclerotic plaque in the contralateral ICA	113 (12,9)	44 (8,8)	28 (7,9)	P1–2: 0,70 P1–3: 0,58 P2–3: 0,9999
SYNTAX Score	15,1±5,2	4,4±2,0	0	P1–2: 0,01 P1–3: 0,001 P2–3: 0,03
ICA clamping time, min	15,1±5,2	4,4±2,0	0	0,1

Table 3

Hospital and long-term complications.

Index	Group 1 (CEE with hemodynamically significant coronary artery stenoses), n=871	Group 2 (CEE with hemodynamically insignificant coronary artery stenoses), n=496	Group 3 (CEA without coronary artery stenosis), n=352	p
Hospital results				
Death	2 (0,22)	2 (0,4)	1 (0,2)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Myocardial infarction	2 (0,22)	1 (0,2)	0	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
ACVA/TIA	1 (0,11)	3 (0,6)	1 (0,2)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Thrombosis of the reconstruction zone	1 (0,11)	0	1 (0,2)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Damage to cranial nerves	145 (16,6)	81 (16,3)	69 (19,6)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Combined endpoint	5 (0,5)	6 (1,2)	2 (0,5)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Long-term results				
Observation period, months	56,8±23,2	62,0±15,6	58,1 ±20,4	0,55
Death from cardiovascular causes	38 (4,3)	20 (4,0)	11 (3,1)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Myocardial infarction (non-fatal)	54 (6,1)	41 (8,2)	36 (10,2)	P1-2: 0,9999 P1-3: 0,8485 P2-3: 0,9999
Repeat coronary revascularization	35 (4,0)	28 (5,6)	25 (7,1)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
ACVA/TIA (non-lethal)	61 (7,0)	35 (7,0)	42 (11,9)	P1-2: 0,9999 P1-3: 0,5728 P2-3: 0,7230

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Table 3 (continued)

Index	Group 1 (CEE with hemodynamically significant coronary artery stenoses), n=871	Group 2 (CEE with hemodynamically insignificant coronary artery stenoses), n=496	Group 3 (CEA without coronary artery stenosis), n=352	p
Revascularization of the ICA from the contralateral side	38 (4,3)	19 (3,8)	27 (7,6)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Hemodynamically significant restenosis in the reconstruction area (more than 60 %)	24 (2,7)	16 (3,2)	13 (3,6)	P1-2: 0,9999 P1-3: 0,9999 P2-3: 0,9999
Combined endpoint	153 (17,5)	96 (19,3)	89 (25,2)	P1-2: 0,9999 P1-3: 0,1966 P2-3: 0,5992

Endpoints

The primary control points were the development of such adverse cardiovascular events as: death, non-fatal myocardial infarction (MI), non-fatal acute cerebrovascular accident/transient ischemic attack (ACVA/TIA), thrombosis/restenosis in the reconstruction area, combined end point (death+ ACVA/TIA+MI).

Statistical analysis

The type of distribution was determined using the Kolmogorov–Smirnov test. Group comparisons were made using Kruskal–Wallis and Pearson χ^2 analyses. Kaplan–Meier analysis was used to construct survival curves. To compare the curves, a Log-rank test was performed. Quantitative data are presented as $M \pm SD$, where M is the mean value, SD is the standard deviation. Qualitative indicators are presented as n (%), where n is the number of patients. Differences were assessed as significant at $p < 0.05$. The research results were processed using the application package GraphPad Prism (www.graphpad.com) and MedCalc 19.2.1 (www.medcalc.org).

Sample characteristics

The groups were comparable in the vast majority of clinical and demographic characteristics and neurological tests. However, among patients with isolated lesions of the ICA, the largest number of female patients was identified. Due to the varying severity of

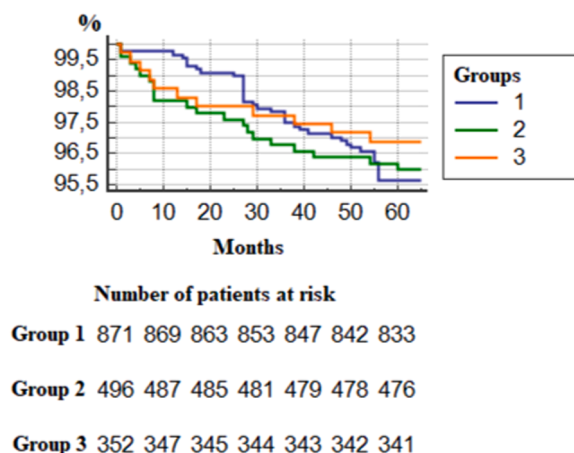


Fig. 1. Freedom from death due to cardiovascular causes.

coronary atherosclerosis, group 1 concentrated the overwhelming number of patients with angina pectoris of functional classes I–II, post-infarction cardiosclerosis, and a history of myocardial revascularization.

According to current recommendations, CEA with minimal risk of complications should be performed in the absence of severe neurological deficit. According to the results of testing of patients, neurological disorders corresponded to a mild degree: 3–8 points on the NIHSS scale; no more than 2 points on the modified Rankin scale; more than 61 points on the Barthel scale (Table 1).

However, the largest number of strokes/TIA in history was recorded in group 3, which is due to the more frequent detection of bilateral ICA stenoses and unstable atherosclerotic plaque among these patients (Table 2).

Results

During the hospital observation period, there were no significant intergroup differences in the number of complications. All cardiovascular events were detected in isolated cases. The most common injury was cranial nerve injury (CNN), diagnosed in every fifth patient in the general sample (Table 3). The composite endpoint was 0.75 % ($n=13$).

In the long-term follow-up period, the groups also turned out to be comparable in the frequency of all existing conditions (Table 3).

However, when comparing survival curves, in group 3 the highest number of stroke/TIA, MI, and combined endpoint was identified. There were no intergroup differences in the number of deaths (Figs. 1–4).

Discussion

Despite the fact that patients with multifocal atherosclerosis (MFA) represent a fairly heterogeneous cohort, the patients in our study were comparable in many ways (Table 1). However, it should be noted that the severity of atherosclerosis directly depended on gender. Severe coronary lesions were more often observed among men, while isolated ICA stenoses developed much more rapidly in women. This trend is known in the literature and is largely associated with hormonal characteristics and estrogen protection.^{16,17} Nevertheless, it is necessary to note the trend towards further rapid progression of coronary and precerebral atherosclerosis in the 3rd group of patients, which is reflected in the increase in the number of adverse ischemic events relative to other samples (Figs. 2–4). Moreover, among the 52 patients of the 3rd group in whom new stenoses were identified in the affected pools, 39 (75 %) were female, 13 (25 %) were male ($p < 0.0001$; OR 9.0; 95 % CI 3.7–21.9); among 78 MI and stroke/TIA of the 3rd group - 51 (65.4 %) were female, 27 (34.6 %) were male ($p = 0.0002$; OR 3.56; 95 % CI 1, 84–6.9). Thus, we can conclude that subclinical and clinical MFA manifests itself later in females, which is associated with a further need for revascularization of new areas and an increase in the number of ischemic complications.

Patients with isolated carotid atherosclerosis had a much more aggravated lesion morphology. These patients are more likely to have bilateral stenoses of the carotid arteries and unstable atherosclerotic plaque, which in turn entailed the formation of the largest number of strokes/TIAs relative to other groups (Table 2). This observation can be explained by the greater reactivity of the local lesion relative to the diffuse process affecting several areas, as well as the cross-effect of inflammatory factors of the operated vessel wall on healthy arteries.^{18–20} However, this nuance did not play a significant role in the indicators of hospital complications. However, upon further observation, it was recorded that the largest number of patients from the total sample with bilateral ICA stenoses were concentrated in group 3. Thus, out of 129 patients, 67 (52 %) received complete cerebral revascularization, and 62 (48 %) did not attend the second stage - CEE on the contralateral side ($p = 0.61$; OR 1.1; 95 % CI 0.71–1.9). The most common reason for refusal was skepticism about the effect of the operation and underestimation of the severity of one's condition. Thus, patients of groups 1 and 2, who had coronary pathology and the largest number of ischemic disasters in their history (Tables 1, 2), were highly compliant relative to group 3, which also contributed to the number of adverse cardiovascular events in the long-term follow-up period (Figs. 2 and 3).

When analyzing survival charts (mortality and MI), the most severe group in terms of the severity of coronary atherosclerosis

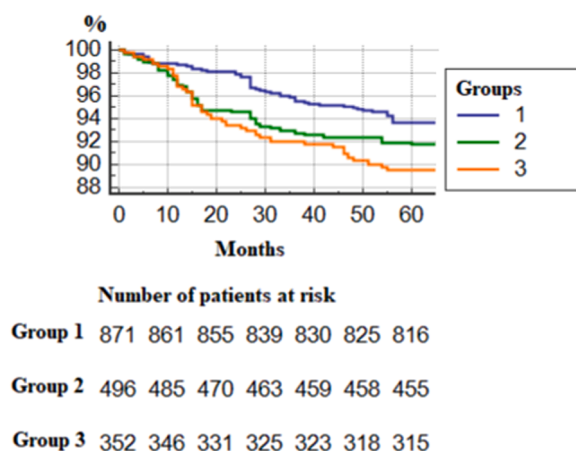


Fig. 2. Freedom from myocardial infarction.

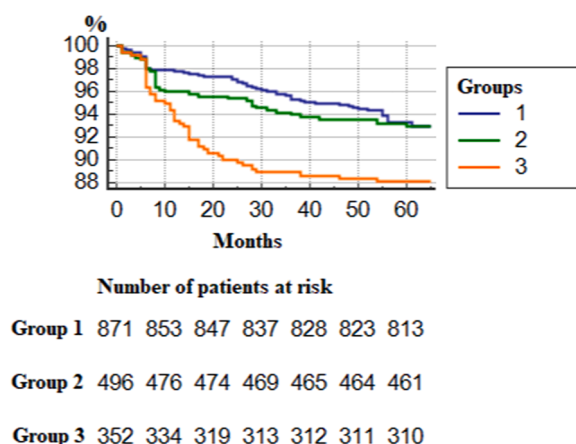


Fig. 3. Freedom from stroke/TIA.

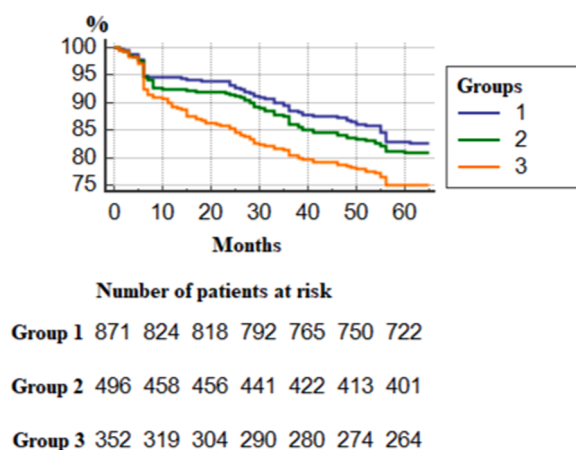


Fig. 4. Freedom from the combined endpoint (death + stroke/TIA + MI).

(group 1) demonstrates optimistic forecasts in the first two years and does not reach the values of groups 2 and 3, which are twice as high in the number of complications. However, after 2 years, between 20 and 30 months, a sharp step is clearly visible on the graph of group 1, which practically equalizes all indicators (Figs. 1 and 2). This fact is associated with the development of restenosis/thrombosis of shunts/stents, which, according to our data, peaks precisely at this period of time.^{21,22} During the same period, more than half of all repeated unplanned myocardial revascularizations of group 1 were recorded.

The largest number of restenoses, occlusions and thromboses of the ICA after classical CEA are diagnosed in the first year of observation, which is most likely due to neointimal hyperplasia as a result of transformation of hemodynamic parameters in the reconstruction zone.^{23,24} Among patients with eversion CEA, loss of vessel lumen occurred predominantly after a year of observation, which confirms the main mechanism of restenosis in the form of progression of atherosclerosis.^{23,24} Moreover, despite the fact that there was no statistically significant difference in the rate of stroke/TIA in the long-term period between the groups, Fig. 3 demonstrates a clear connection between the development of ischemic stroke and the development of restenosis/occlusion/thrombosis, depending on the above conclusions. Over a period of time not exceeding 10 months, a sharp step in all three graphs is determined, reflecting the development of an ischemic catastrophe associated with the formation of restenosis. It is during this period of time that the largest number of ICA lumen losses occurs, which is associated with the overwhelming number of classical CEA in the total sample ($n=1256$; 73 %; $p<0.0001$; OR 7.35; 95 % CI 6.32–8.55).

One of the problems of modern carotid surgery is postoperative cranial neuropathy.^{1,4,6,9,15} Studies have not established the dependence of this complication on the type of CEE.^{1,7,17,19,24} The greatest attention deserves trauma to the glossopharyngeal plexus, the consequences of which are not reversible. In our work, this complication was recorded in 3 cases after classical CEE and was accompanied by impaired swallowing function. All patients developed aspiration pneumonia, which regressed with antibiotic therapy. In two cases, a gastrostomy tube was installed; one patient adapted to the act of swallowing using the other side. A year after CEA, despite intensive neuroprotective, anti-inflammatory, physical therapy, swallowing function was not restored, the patients continued to receive nutrition through a gastrostomy tube.

Conclusion

The results of the study showed that there was no significant intergroup difference in the development of complications at the hospital postoperative stage. However, in the long-term follow-up period, a group of patients with isolated lesions of the ICA demonstrated a rapid increase in the number of MI, stroke/TIA, and a combined endpoint, which was apparently associated with low compliance and progression of atherosclerosis in previously unaffected arteries.

Funding

No funding was received for the study.

Informed consent

Informed consent was obtained from the patients.

Ethical conclusion

The study was performed in compliance with the ethical principles of conducting scientific medical research involving humans and was approved by the decision of the local ethical committee of the Federal State Budgetary Institution of Higher Education "Amur State Medical Academy" of the Ministry of Health of the Russian Federation (protocol No. 9 dated August 13, 2023). The study was performed in accordance with the Good Clinical Practice standards and the principles of the Declaration of Helsinki, did not contradict the Federal Law of the Russian Federation of November 21, 2011 No. April 1, 2016 N 200n "On approval of the rules of good clinical practice".

Declaration of Competing Interest

None in the template.

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